

Basics of Woodgas

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An overview by Steve Unruh

So here is my current evolved explanation: Wood gasification is running smoke through a hot bed of charcoal.

Period. That's it.

Of course, the devil is in the details. You have to set things up so ONLY this is happening. I answer people's questions by referring back to these statements. Example: Have you ever burned wood chips? I have. Smokes a lot. But how you gonna flow that smoke through a packed bed of charred chips?

What about firewood? You have to create a good thick bed of hot, hot glowing char chunks. Any gaps and the smoke will flow through unconverted and tar up/stick your intake valves. Think you can do this? Some of these folk can. I can. Short, hand stacked batch runs. The efficiency really sucks though - it produces more heat than usable gas, using too much air resulting in too much Nitrogen carry through and too much produced CO₂ in the gas. Much better to chunk up the wood to the proper size for a gasifier hearth.

Basics of Gasification

by Jim Mason

Gasification is most simply thought of as a process of staged or choked combustion. It is burning solid fuels like wood or coal without enough air to complete combustion, so the output gas still has combustion potential. The gas produced by this method goes by a variety of names: "wood gas", "syngas", "producer gas", "town gas", "generator gas", and others. It's sometimes also called "biogas", though biogas more typically refers to gas produced via microbes in anaerobic digestion.

You might think of gasification as burning a match, but interrupting the process by piping off the clear gas you see right above the match, not letting it mix with oxygen and complete combustion. Or you might think of it as running your car engine extremely rich, creating enough heat to break apart the raw fuel, but without enough oxygen to complete combustion, thus sending burnable gasses out the exhaust. This is how a hot rodder gets flames out the exhaust pipes.

The input to gasification is some form of solid carbonaceous material— typically biomass or coal. All organic carbonaceous material is made up of carbon (C), hydrogen (H), and oxygen (O) atoms— though in a dizzying variety of molecular forms. The goal in gasification is to break down this wide variety of forms into the simple fuel gasses of H₂ and CO— hydrogen and carbon monoxide.

Both hydrogen and carbon monoxide are burnable fuel gasses. We do not usually think of carbon monoxide as a fuel gas, but it actually has very good combustion characteristics (despite its poor characteristics when interacting with human hemoglobin). Carbon monoxide and hydrogen have about the same energy density by volume. Both are very clean burning as they only need to take on one oxygen atom, in one simple step, to arrive at the proper end states of combustion, CO₂ and H₂O. This is why an engine run on syngas can have such clean emissions. The engine becomes the “afterburner” for the more dirty and difficult early stages of combustion that now are handled in the gasifier.

The 4 processes of gasification.

Now let's complicate things slightly. “Proper” gasification is a bit more than just the “choked combustion” summary above. It is actually a series of distinct thermal events put together so as to purpose convert solid organic matter into specific hydrocarbon gasses as output. Simple incomplete combustion is a dirty mess. The goal in gasification is to take control of the discrete thermal processes usually mixed together in combustion, and reorganize them towards desired end products. In digital terms, “Gasification is the operating system of fire”. Once you understand its underlying code, you can pull fire apart and reassemble it to your will, as well as a stunning variety of end products and processes.

Gasification is made up for 4 discrete thermal processes: Drying, Pyrolysis, Combustion and Reduction. All 4 of these processes are naturally present in the flame you see burning off a match, though they mix in a manner that renders them invisible to eyes not yet initiated into the mysteries of gasification. Gasification is merely the technology to pull apart and isolate these separate processes, so that we might interrupt the “fire” and pipe the resulting gasses elsewhere.

Two of these processes tend to confuse all newcomers to gasification. Once you understand these two processes, all the others pieces fall in place quickly. These two non-obvious processes are Pyrolysis and Reduction. Here's the quick cheat sheet.

Pyrolysis: Pyrolysis is the application of heat to raw biomass, in an absence of air, so as to break it down into charcoal and various tar gasses and liquids.

Biomass begins to “fast decompose” with once its temperature rises above around 240C. The biomass breaks down into a combination of solids, liquids and gasses. The solids that remain we commonly call “charcoal”. The gasses and liquids that are released we collectively call “tars”.

The gasses and liquids produced during lower temp pyrolysis are simply fragments of the original biomass that break off with heat. These fragments are the more complicated H, C and O molecules in the biomass that we collectively refer to as volatiles. As the name suggests, volatiles are “reactive”. Or more accurately, they are less strongly bonded

in the biomass than the fixed carbon, which is the direct C to C bonds.

Thus in review, pyrolysis is the application of heat to biomass in the absence of air/oxygen. The volatiles in the biomass are “evaporated” off as tars, and the fixed carbon-to-carbon chains are what remains— otherwise known as charcoal.

Reduction: Reduction is the process stripping of oxygen atoms off completely combusted hydrocarbon (HC) molecules, so as to return the molecules to forms that can burn again. Reduction is the direct reverse process of combustion. Combustion is the combination of an HC molecule with oxygen to release heat. Reduction is the removal of oxygen from an HC molecule by adding heat. Combustion and Reduction are equal and opposite reactions. In fact, in most burning environments, they are both operating simultaneously, in some form of dynamic equilibrium, with repeated movement back and forth between the two states.

Reduction in a gasifier is accomplished by passing carbon dioxide (CO₂) or water vapor (H₂O) across a bed of red hot char (C). The hot char is highly reactive with oxygen, and thus strips the oxygen off the gasses, and redistributes it to as many single bond sites as possible. The oxygen is more attracted to the bond site on the C than to itself, thus no free oxygen can survive in its usual diatomic O₂ form. All available oxygen will bond to available C sites as individual O, until all the oxygen is gone. When all the available oxygen is redistributed as single atoms, reduction stops.

Through this process, CO₂ is reduced to CO. And H₂O is reduced to H₂ and CO. Combustion products become fuel gasses again. And those fuel gasses can then be piped off to do desired work elsewhere.

Combustion and Drying: These are the most easily understood of the 4 Processes of Gasification. They do what we think by common understanding, though now they do it in the service of Pyrolysis and Reduction.

Combustion is what generates the heat to run reduction, as well as the CO₂ and H₂ to be reduced in Reduction. Combustion can be fueled by either the tar gasses or char from Pyrolysis. Different reactor types use one or the other or both. In a downdraft gasifier, we are trying to burn the tar gasses from pyrolysis to generate heat to run reduction, as well as the CO₂ and H₂O to reduce in reduction. The goal in combustion in a downdraft is to get good mixing and high temps so that all the tars are either burned or cracked, and thus will not be present in the outgoing gas. The char bed and reduction contribute a relatively little to the conversion of messy tars to useful fuel gasses. Solving the tar problem is mostly an issue of the reaction dynamics in the combustion zone.

Drying is what removes the moisture in the biomass before it enters Pyrolysis. All the moisture needs to be (or will be) removed from the fuel before any above 100C processes happen. All of the water in the biomass will get vaporized out of the fuel at some point in the higher temp processes. Where and how this happens is one of the major issues that has to be solved for successful gasification. High moisture content fuel, and/or poor handling of the moisture internally, is one of the most common reasons for failure to produce clean gas.

About Drive On Wood

A community of gasification enthusiasts hosted by Wayne Keith, resident expert & woodgas pioneer.

Drive On Wood is a partnership of Wayne Keith & Chris Saenz.

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